

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**8th Semester of B. Tech (ME) Examination****May 2018****ME 409.01/ME 409 Alternate Energy Sources****Date: 04.05.2018, Friday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Answer the question below. [07]**
- i. Discuss basic principle of photovoltaic system or solar cells. [02]
 - ii. Define solar azimuth angle and solar zenith angle. [02]
 - iii. Solar cell are made from the following materials [01]
 - (a) crystalline (b) ferrite
 - (c) iron (d) silver
 - iv. When the water is ejected from earth's interior in the form of hot water, it is called [01]
 - (a) Geyser (b) Hot springs
 - (c) Both (a) and (b) (d) None of the above
 - v. Identify the non-renewable energy resource from the following: [01]
 - (a) Coal (b) Fuel cells
 - (c) Wind power (d) Wave power
- Q – 2.a Write short note on classification of energy sources. [04]**
- Q – 2.b Answer any two questions. [10]**
- (i) Calculate number of day light hours at Delhi on Jan 21st and June 21st in year 2013. Take latitude for Delhi 28°35'
 - (ii) What are the main components of flat plate solar collector, explain the function of each.
 - (iii) Describe the closed cycle OTEC power plant and state its advantages over open cycle system.
- Q - 3 Answer any two questions. [14]**
- a. What is the solar still? Explain it in detail with sketch.
 - b. Explain construction and working of pyrheliometer with sketch.
 - c. Write short note on savonious rotor and darrius rotor.

SECTION – II

Q - 4 Answer the question below.

[07]

- i. The bio-gasification is a process also referred as
 - (a) aerobic digestion
 - (b) anaerobic digestion
 - (c) fermentation
 - (d) thermochemical gasification
- ii. The fluid that cannot be used in OTEC system as working fluid is
 - (a) ammonia
 - (b) liquid sodium
 - (c) refrigerating fluid
 - (d) steam
- iii. The efficiency of an OTEC system is about
 - (a) 5%
 - (b) 20%
 - (c) 30 %
 - (d) 40%
- iv. In anaerobic digestion, the useful end product is,
 - (a) carbon dioxide
 - (b) hydrogen
 - (c) methane
 - (d) Freon
- v. The minimum wind velocity required for wind turbine is called
 - (a) cut-in speed
 - (b) cut-out speed
 - (c) rated speed
 - (d) tip speed
- vi. The output power delivered by an MHD generator is proportional to
 - (a) (magnetic flux density)²
 - (b) (magnetic flux density)
 - (c) (magnetic flux density)⁻²
 - (d) (magnetic flux density)⁻¹
- vii. Define solar constant.

Q – 5.a Design the rotor radius for multiblade wind turbine that operates in a wind speed of 26 km/hr to pump water at a rate of 16 m³/h with a lift of 15m. Also, calculate the angular velocity of the rotor. The following data is given : Water density is 1000 kg/m³, g = 9.81 m/s, Water pump efficiency = 60%, transmission efficiency of rotor to pump = 90%, power coefficient = 0.3, tip speed ratio = 0.8, and air density = 1.18 kg/m³ **[05]**

OR

Q – 5.a Explain the working of Dolphin type wave energy converter. **[05]**

Q – 5.b Explain site selection for OTEC. **[04]**

Q – 5.c Derive an equation for maximum power, maximum torque and maximum axial thrust available from wind turbine. **[05]**

OR

Q – 5.c With the suitable schematic diagram explain the working for solar water heating. **[05]**

Q – 6.a Describe working of Janta biogas plant briefly with figure. Also write its demerits. **[07]**

Q – 6.b Describe the binary cycle geothermal power plant. **[07]**

OR

Q – 6.a Classify gasifiers and explain downdraft type gasifier with neat sketch. **[07]**

Q – 6.b Explain with figure basic working principle of MHD generator and state its advantages and disadvantages. **[07]**
